



IBM Developer  
SKILLS NETWORK

# Winning Space Race with Data Science

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# Outline

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- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

# Executive Summary

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- Summary of methodologies
- Summary of all results

# Introduction

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- Goal: To make space travel more affordable by reusing rockets
- Costsavings:Falcon9launch = \$62m vs \$165m+ (competitors)
- KeyFactor:Successfulfirst-stagelandingreduceslaunchcosts
- Objective: Predict if SpaceX will reuse the first stage based on public data
- Key Questions:
  - How do payload mass, launch site, flights, and orbits impact success?
  - Has the landing success rate improved over time?
  - What is the best algorithm for this binary classification problem?

Section 1

# Methodology

# Methodology

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## Executive Summary

- Data collection methodology:
  - Data collection: SpaceX REST API and Wikipedia web scraping
- Perform data wrangling
  - Filtering , handling missing values, one hot encoding
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Predictive Modeling: Building , tuning & evaluating classification models

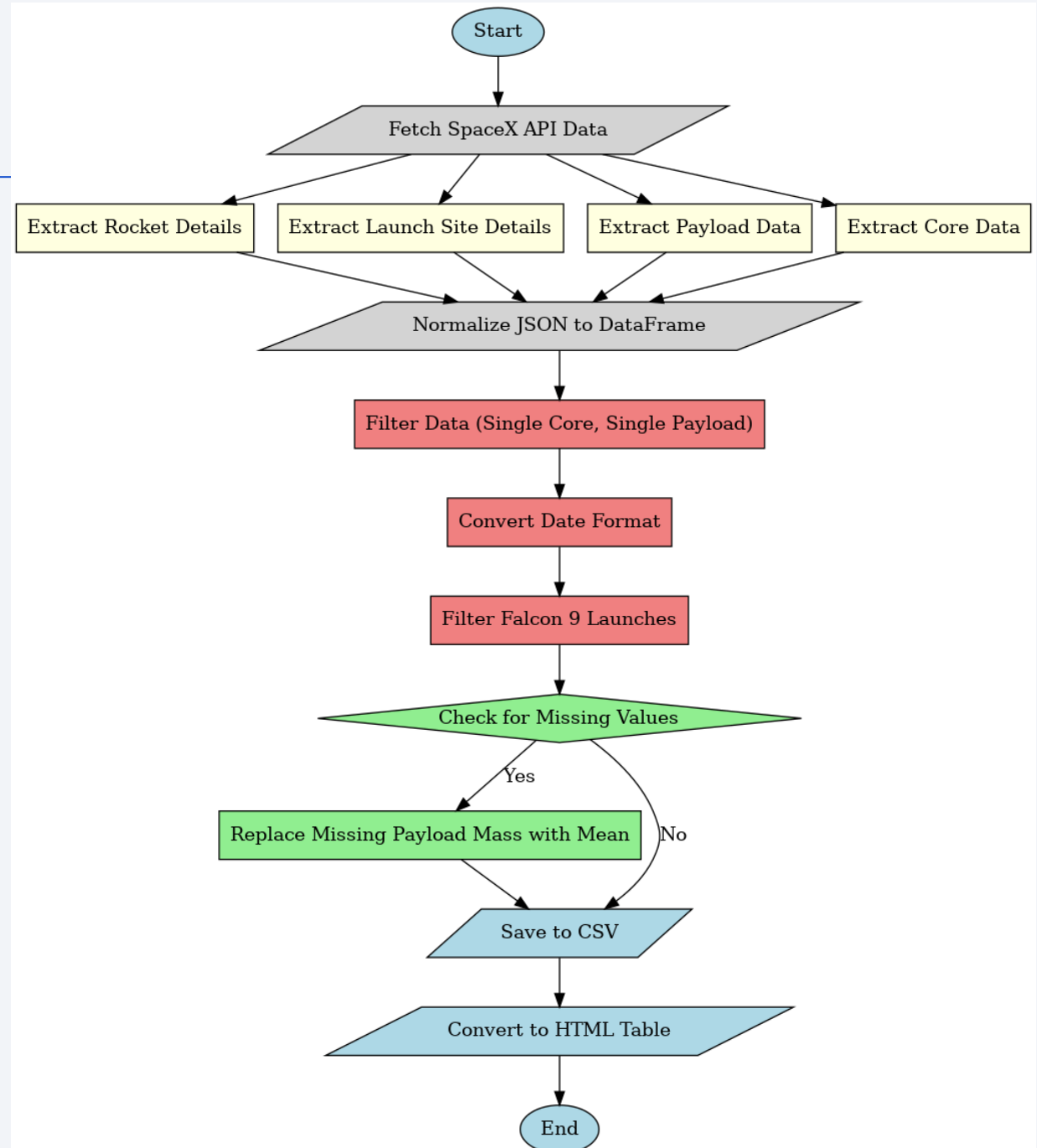
# Data Collection

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- Sources:
  - SpaceX API (structured launch data)
  - Wikipedia (historical launch records via web scraping)
- Process:
  - API: Extract, normalize, clean, and preprocess data
  - Web Scraping: Parse HTML, extract tables, clean & structure data
- Output:
  - Processed datasets stored as CSV and HTML tables for further analysis.

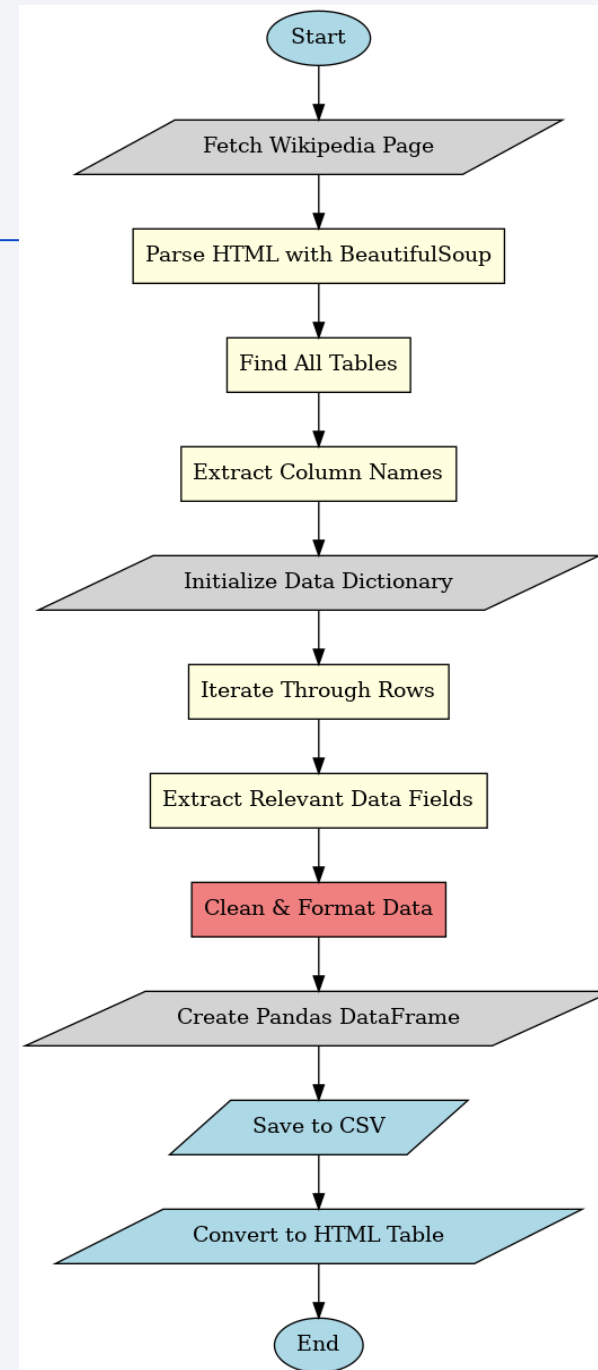
# SpaceX API

- [https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/Data\\_collection\\_API.py](https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/Data_collection_API.py)



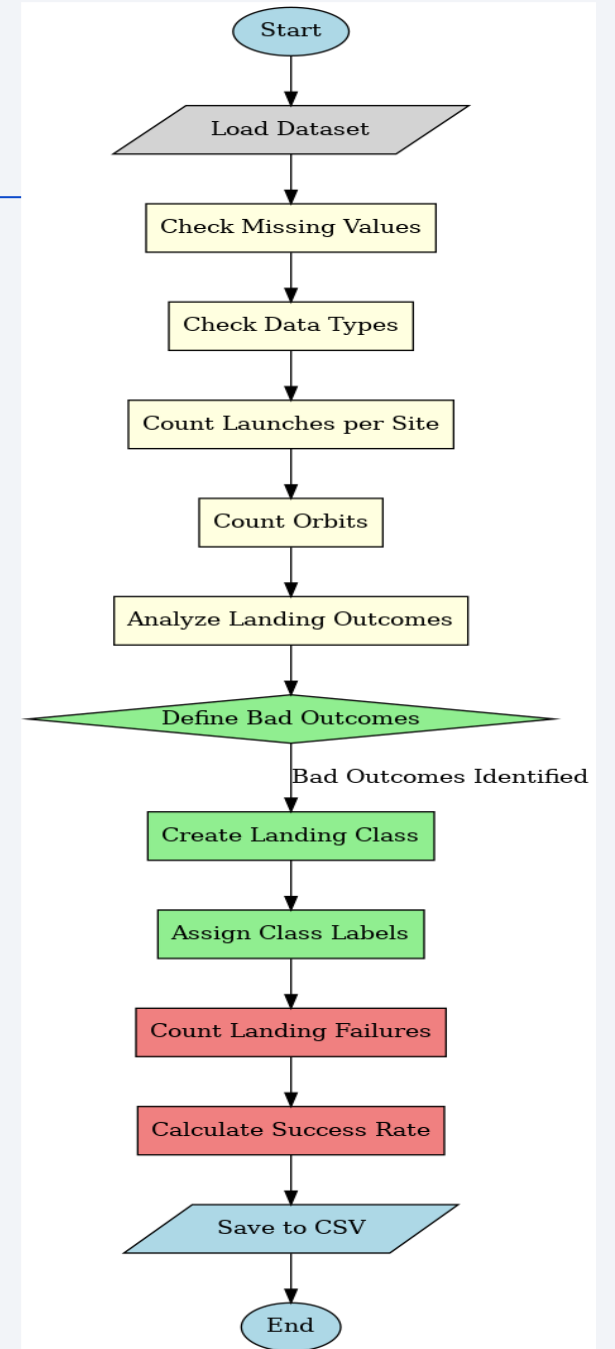
# Data Collection - Scraping

- <https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/web scraping.py>



# Data Wrangling

- Goal: Identify patterns in the dataset to define training labels.
- Booster Landing Outcomes: Successful or unsuccessful
- Approach:
  - Analyze historical launch data
  - Identify key factors influencing booster landings
  - Prepare data for predictive modeling
- [https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/data\\_wrangling.py](https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/data_wrangling.py)



# Build an Interactive Map with Folium

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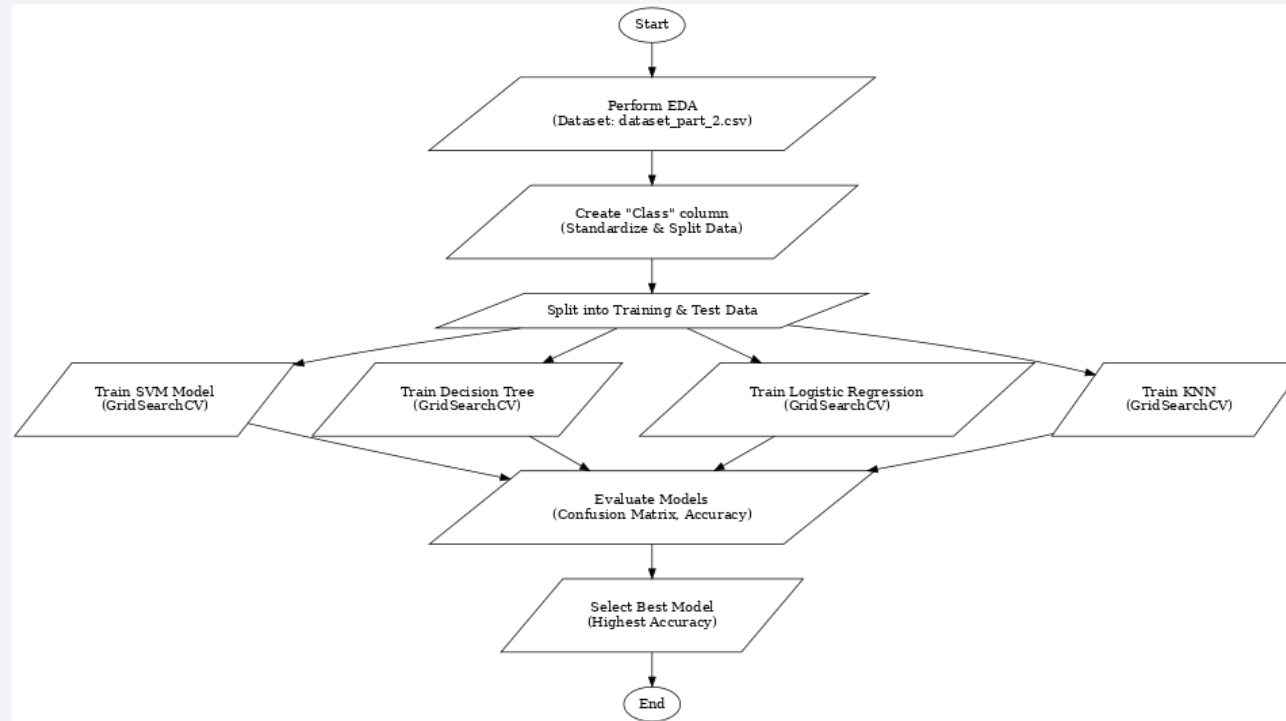
- Does rocket launch success depend on launch site location?
- Launch Site Analysis
  - All sites mapped with coordinates and success classification (green = success, red = failure)
- Proximity Analysis of Launch Sites
  - Are all sites placed at a safe distance away from cities ?
  - Are all launch sites close to the coast ?
  - Are all launch sites close to railway and/or highway stations ?
- <https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/Locations.py>

# Build a Dashboard with Plotly Dash

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- SpaceX Launch Data Analysis w. Interactive plots and components added:
  - Launch Site Dropdown & Payload Range Slider:
    - Plot success rate pie chart
    - Plot scatter plot to show relationship between payload mass and launch outcomes
    - Enable real-time data filtering, making the dashboard interactive and insightful
    - Identifies trends between payload weight and mission success
- [https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/dash\\_interactivity.py](https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/dash_interactivity.py)

# Predictive Analysis (Classification)

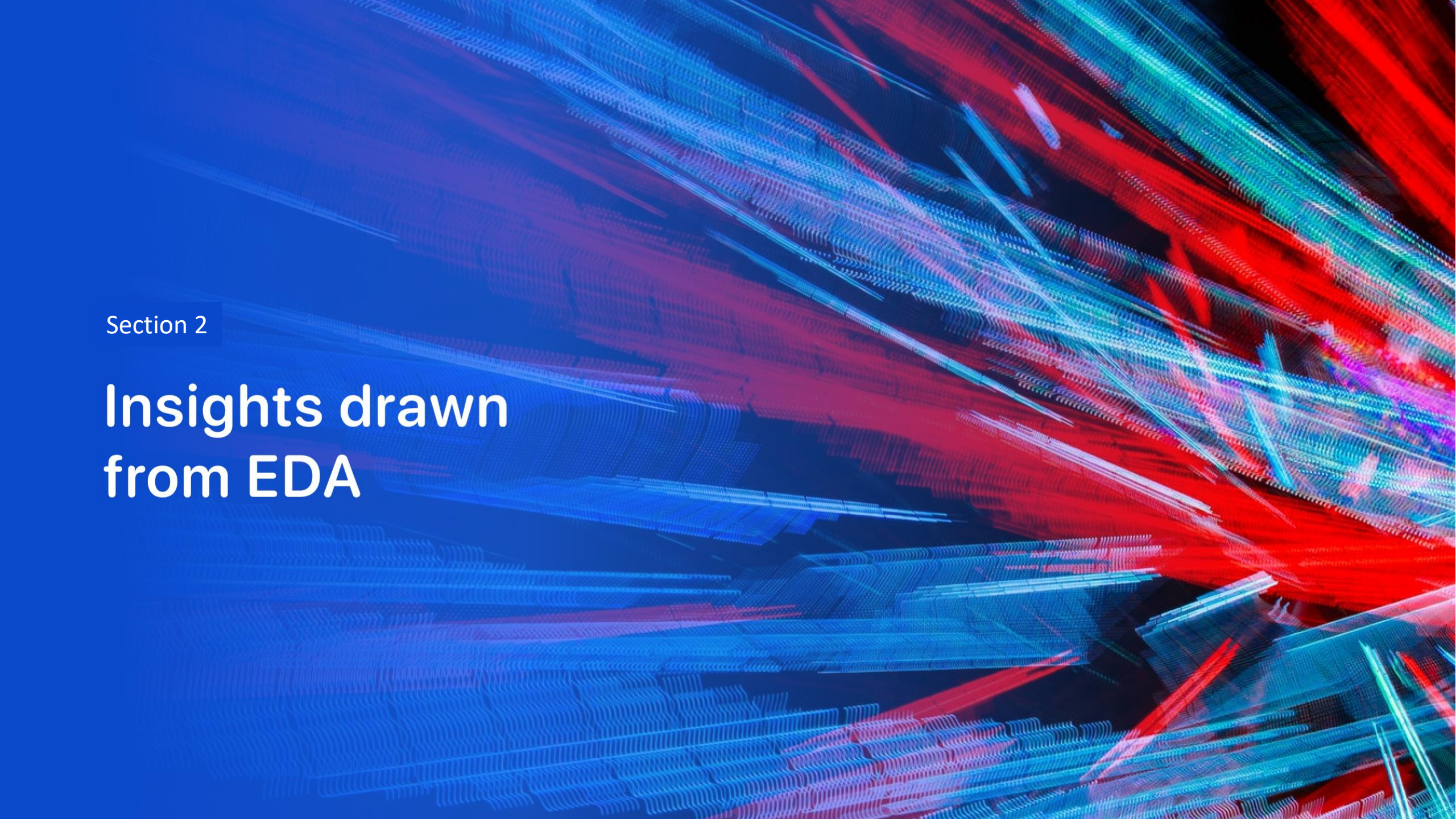


- <https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/ML.py>

# Results

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- Exploratory data analysis results
  - study relationships between variables (scatterplots), compare measurements from discrete categories (bar plots), and study trends over time (line charts). SQL queries focused on launch site usage, distributions of payload masses, landing success/failure rates and how these outcomes can be categorized).
- Interactive analytics demo in screenshots
  - launch site drop down list, pie charts with success launches, payload mass range slider and a scatter chart).
- Predictive analysis results

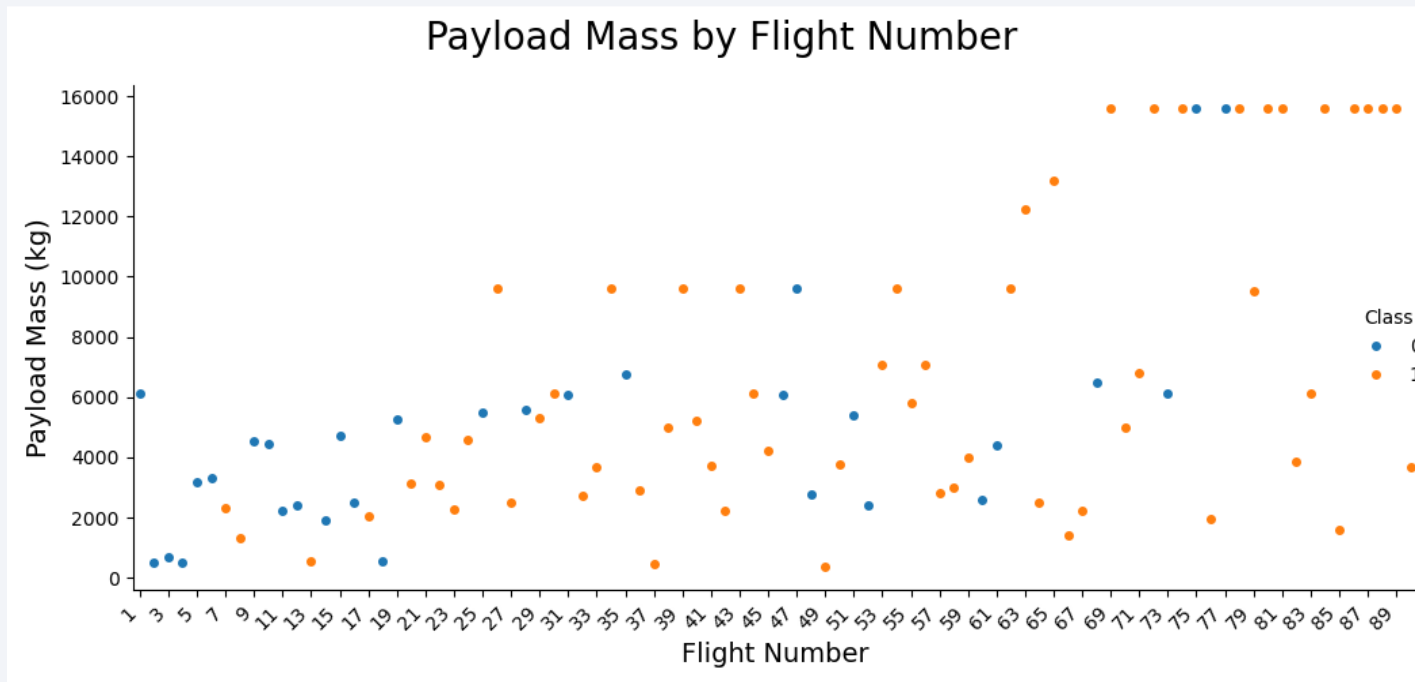
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

# Insights drawn from EDA

# Flight Number vs. Payload Mass

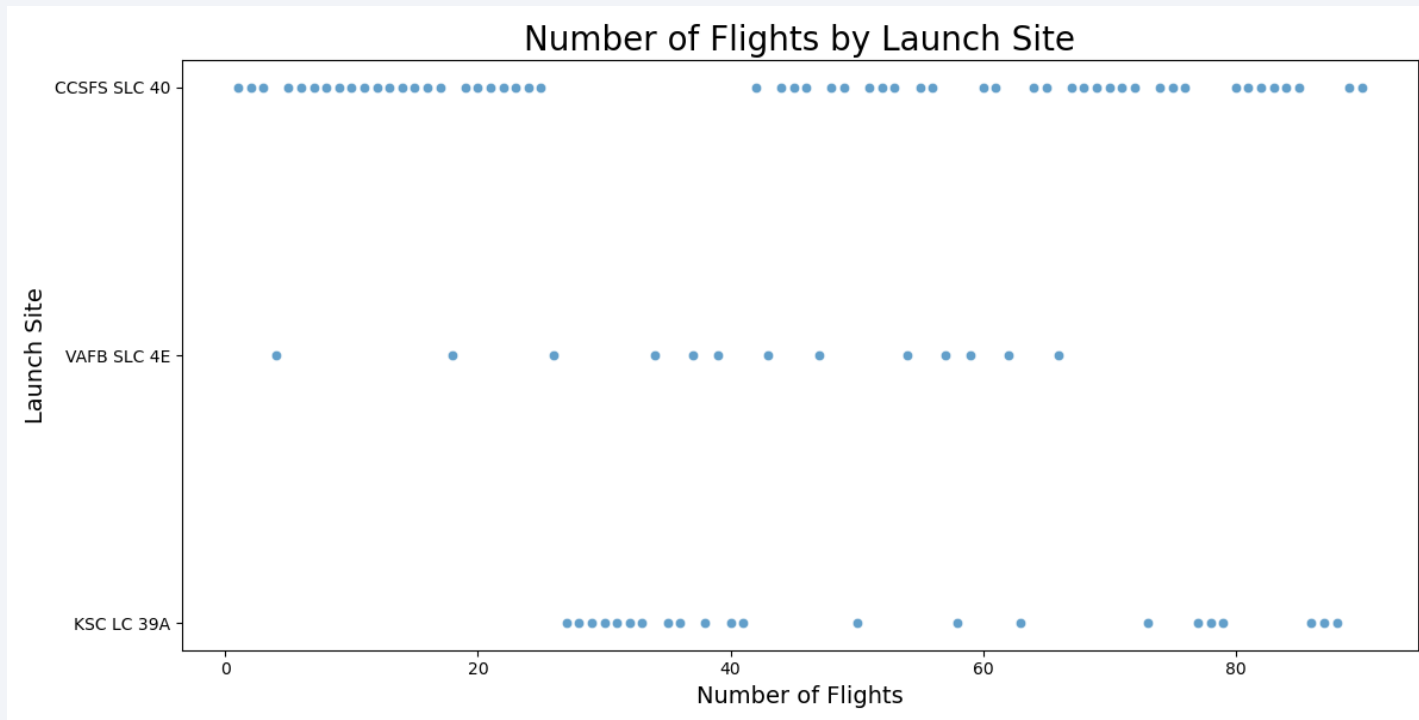
- The learning curve of SpaceX Falcon 9 missions
- Increasing reliability over time and higher payload capabilities in later missions.



## Payload Mass Trends

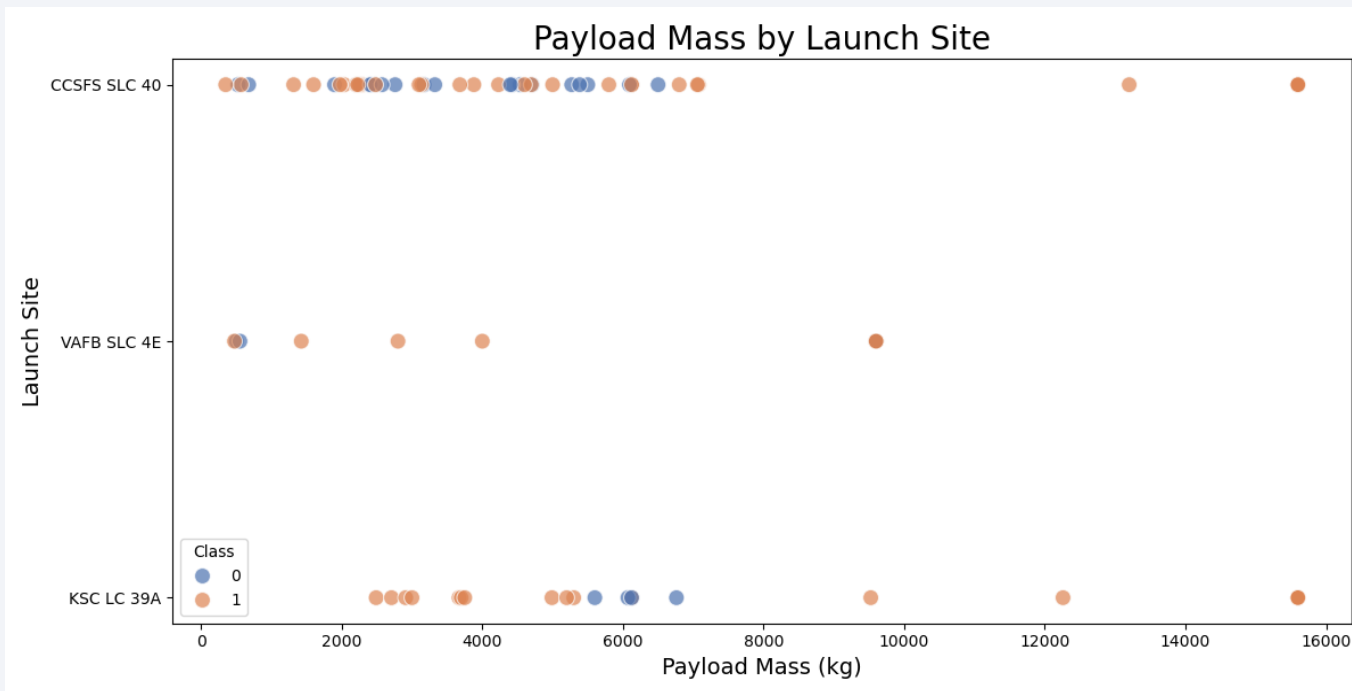
- Higher payload missions have higher success rate in later flights.
- Earlier flights show a mix of classes , while later flights are mostly class 1, which indicates
- improved performance over time
- Around flight number 60 and beyond there is a cluster of higher payloads (16000kg), which seem to be mostly successful

# Payload vs. Launch Site



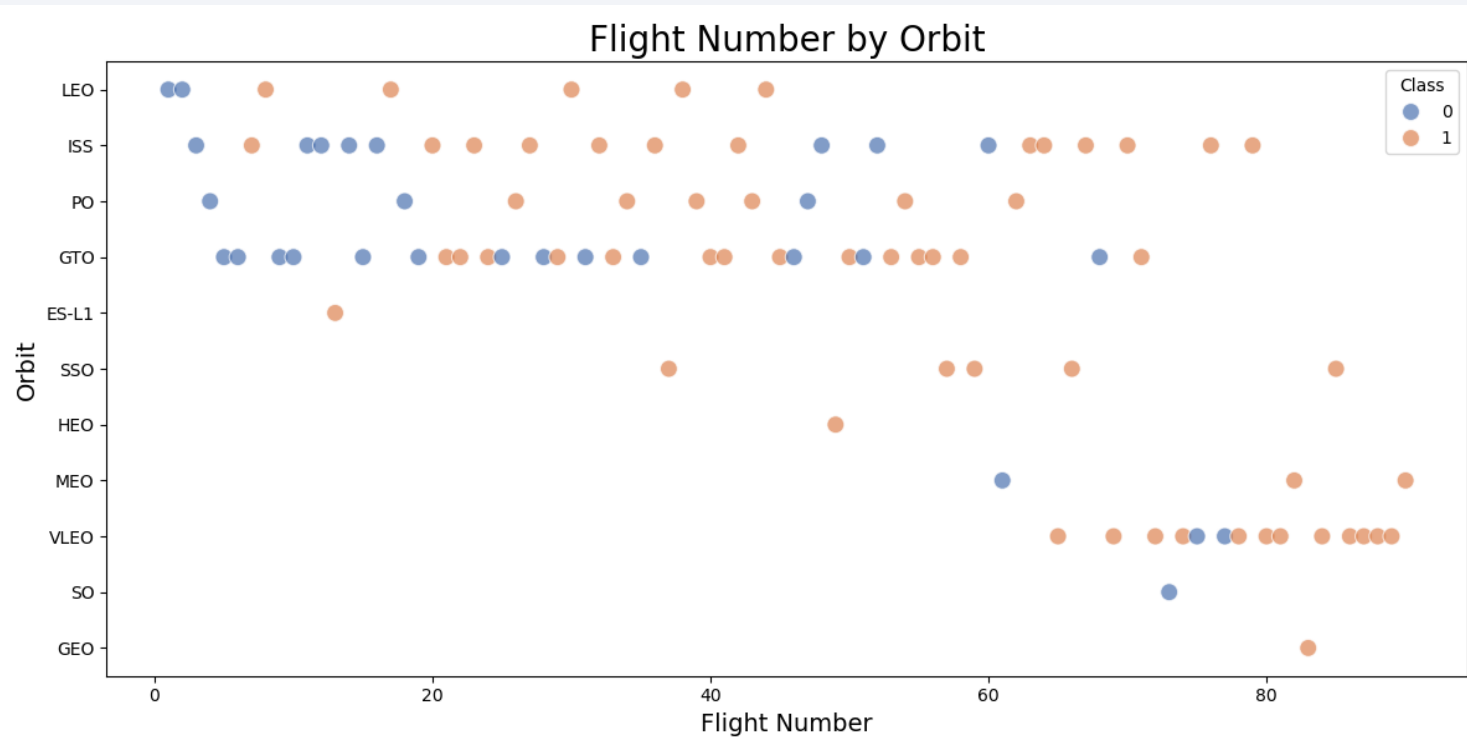
- Distribution of flight across sites.
- CCSFS&KSC LC 39A appear to be primary launch sites.
- CCSFS SLC 40 has the highest number of flights.
- VAFB SLC 4E has significantly fewer flights compared to the other sites.
- → Check specific requirements and logistical constraints as possible explanations.

# Payload vs. Launch Site



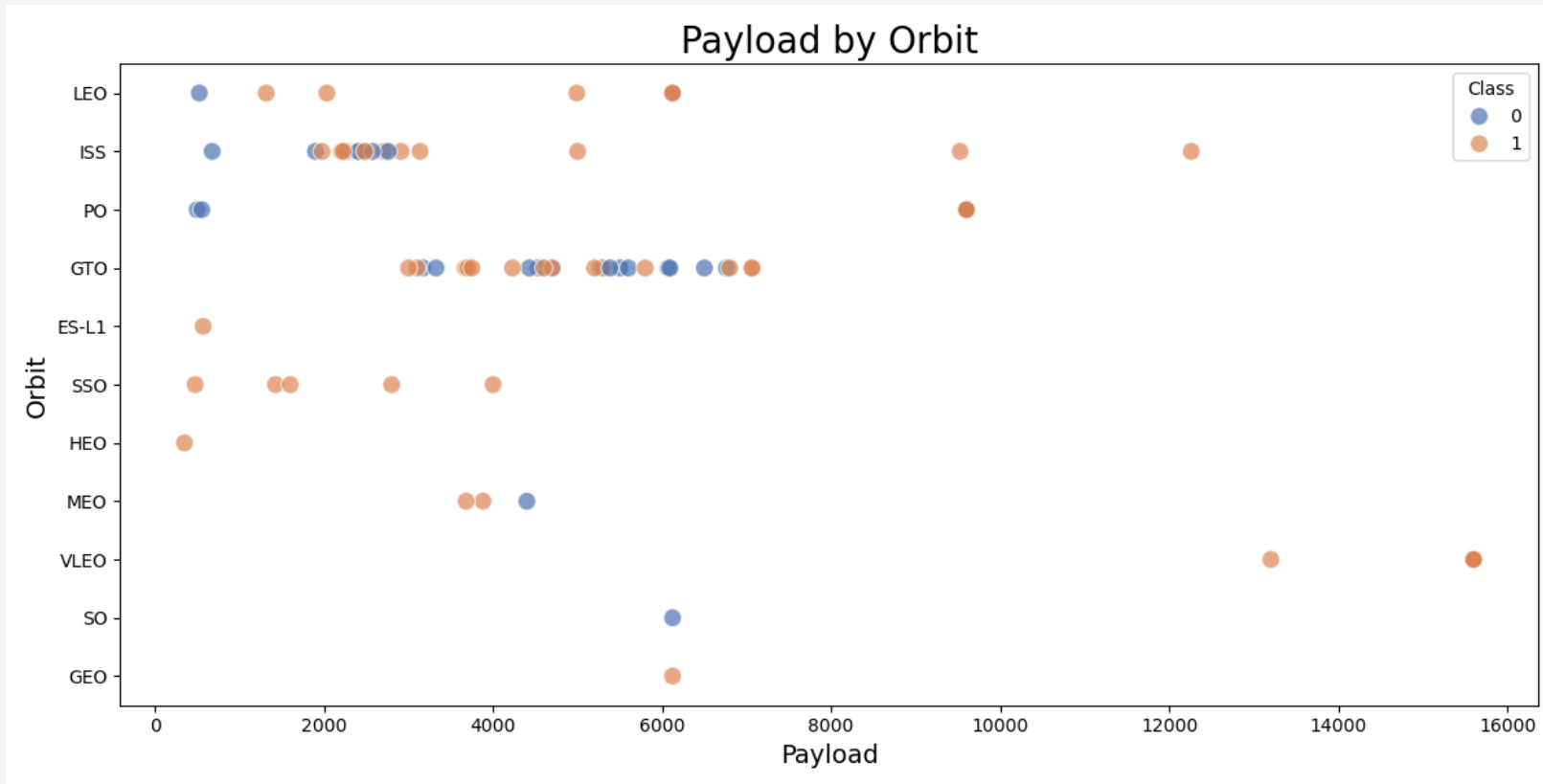
- CCSFSSLC40 is the most used site, supporting a broad range of payloads. VAFBSLC4E is the least used site.
- KSCLC39A has the highest concentration of successful landings.
- Most payloads are in the 0–6,000 kg range across sites.
- Higher payloads (8,000+ kg) appear mostly at CCSFSSLC40 & KSCLC39A and tend to have more successful landings.

# Flight Number vs. Orbit Type



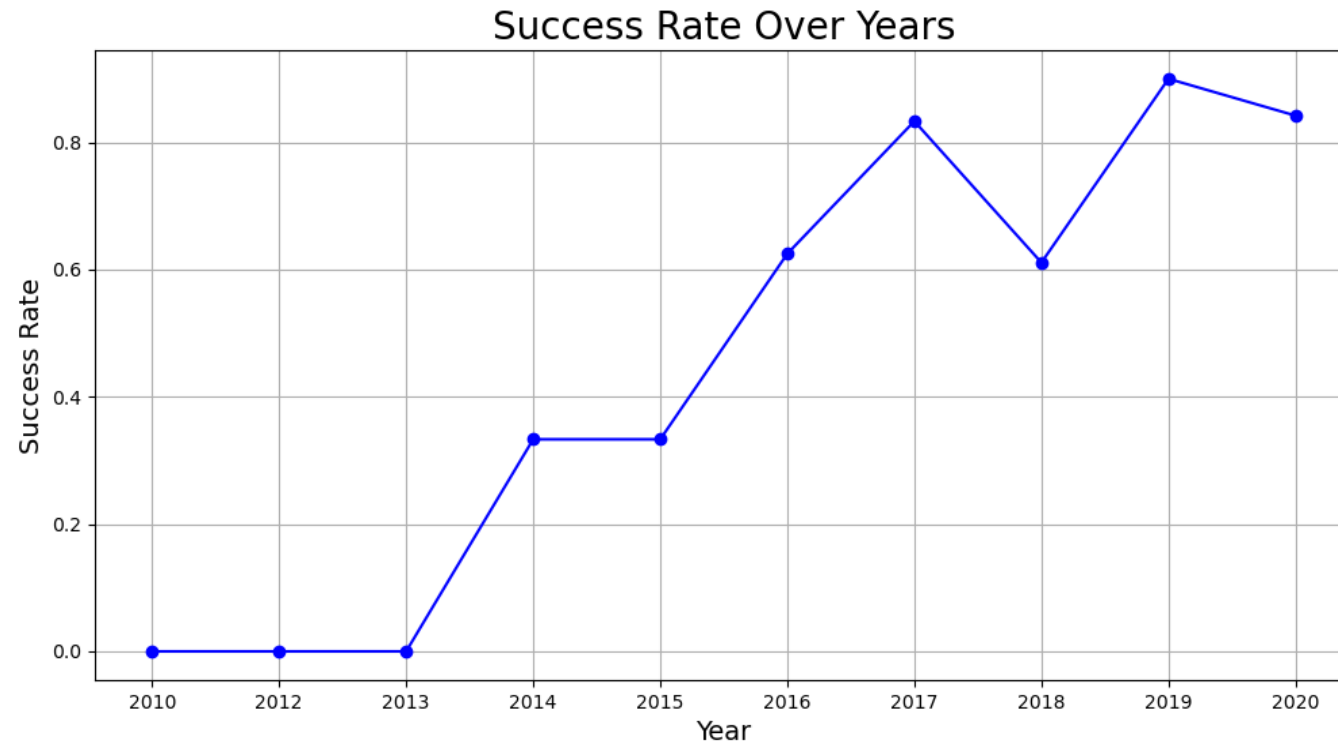
- Mission reliability has significantly improved over time.
- Later missions consistently outperforms earlier ones in success rates.
- Reinforces that GTO remains challenging despite many flights.
- Highlights experience improvements for LEO.
- SO has no success data.

# Payload vs. Orbit Type



- LEO & ISS: Higher payloads tend to succeed more often, while smaller payloads experience more failures.
- GTO shows mixed success rate.
- The correlation between payload mass and success rate is weak →
- success rate does not strongly depend on payload mass.

# Launch Success Yearly Trend



Early failures (2010-2013) reflect an experimental phase, possibly with early rocket iterations.

2014-2015 shows improvements, potentially from lessons learned and technical refinements.

2016-2017 marks a breakthrough period, with dramatic reliability gains.

2018's dip suggests new risks, possibly testing advanced payloads or new technologies.

2019-2020 represents a well-established and reliable system, likely benefiting from improved engineering, automation, and launch procedures.

# All Launch Site Names

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- # Query to display unique launch sites

```
cur.execute('SELECT DISTINCT Launch_site FROM spacex')
```

```
unique_launch_sites = cur.fetchall()
```

```
print("Unique Launch Sites:")
```

```
for site in unique_launch_sites:
```

```
    print(site[0])
```

- Unique launch sites identified:
- CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40

<https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/EDA.py>

# Launch Site Names Begin with 'CCA'

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- Records of Launch sites beginning with '

('2010-06 04', '18:45:00', 'F9 v1.0 B0003', 'CCAFS LC 40', 'Dragon Spacecraft Qualification Unit'

('2010-12 08', '15:43:00', 'F9 v1.0 B0004', 'CCAFS LC 40', 'Dragon demo flight C1, two CubeSats'

('2012-05 22', '7:44:00', 'F9 v1.0 B0005', 'CCAFS LC 40', 'Dragon demo flight C2', 525, 'LEO (ISS)'

('2012-10 08', '0:35:00', 'F9 v1.0 B0006', 'CCAFS LC 40', 'SpaceX CRS 1', 500, 'LEO (ISS)', 'NASA'

('2013-03 01', '15:10:00', 'F9 v1.0 B0007', 'CCAFS LC 40', 'SpaceX CRS 2', 677, 'LEO (ISS)', 'NASA'

# Total Payload Mass

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- Total Payload Mass by NASA (CRS): 45,596 kg
- NASA's CRS (Commercial Resupply Services) missions have transported a total of 45,596 kg of payload to space.
- These missions primarily resupply the International Space Station (ISS) with cargo, equipment, and scientific instruments.

# Average Payload Mass by F9 v1.1

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- The Falcon 9 v1.1 booster had an average payload mass of 2,928.4 kg per launch.
- This version of the Falcon 9 was an upgraded variant, used from 2013 to 2016, featuring higher thrust and increased payload capacity compared to its predecessor.

# First Successful Ground Landing Date

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- The first-ever successful landing of a SpaceX booster on a ground pad occurred on December 22, 2015.
- This historic event took place at Landing Zone 1 (LZ-1) at Cape Canaveral, marking the first time a reusable orbital-class rocket successfully returned to Earth and landed vertically.
- The booster involved in this landing was from the Falcon 9 Full Thrust version, paving the way for SpaceX's reusability strategy in rocket launches.

## Successful Drone Ship Landing with Payload between 4000 and 6000

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- Boosters with successful drone ship landings and payload mass between 4000 and 6000 kg:  
F9 FT, F9 B5
- F9FTB1022,F9FTB1026,F9FTB1021.2,F9FTB1032.1,F9B4B1040.1,F9FTB1031.2,F9B4B1043.1,F9B5B1046.2,F9B5B1047.2,F9B5B1046.3,F9B5B1048.3,F9B5B1051.2,F9B5B1060.1,F9B5B1058.2,F9B5B1062.1
- The Falcon 9 Full Thrust (F9 FT), Falcon 9 Block 4 (F9 B4), and Falcon 9 Block 5 (F9 B5) boosters successfully landed on drone ships while carrying payloads between 4000 and 6000 kg.
- This demonstrates SpaceX's progress in booster reusability, enabling successful recoveries even with heavier payloads

# Total Number of Successful and Failure Mission Outcomes

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- 61 successful landings indicate that SpaceX has a high success rate in booster recovery.
- 10 failed landings show that while there were challenges, SpaceX has steadily improved its landing technology and reliability over time.
- These numbers highlight the effectiveness of reusable rocket technology, reducing costs and increasing the frequency of launches.

# Boosters Carried Maximum Payload

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- Booster Versions with Maximum Payload Mass: F9 B5
- B1048.4, F9B5B1049.4, F9B5B1051.3, F9B5B1056.4, F9B5B1048.5, F9B5B1051.4, F9B5B1049.5, F9B5B1060.2, F9B5B1058.3, F9B5B1051.6, F9B5B1060.3, F9B5B1049.7
- All the boosters listed belong to the Falcon 9 Block 5 (F9B5) series, which is SpaceX's most powerful and reusable booster variant.
- These boosters carried the heaviest payloads recorded in SpaceX missions, showcasing F9B5's superior thrust and lifting capability.
- The reuse of specific boosters (e.g., B1049, B1051, B1060) multiple times proves the effectiveness of SpaceX's reusability strategy.

# 2015 Launch Records

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- Records with failure landing outcomes in 2015:
  - ('CCAFS LC-40', 'F9 v1.1 B1012', 'Failure', '2015-01-10')
  - ('CCAFS LC-40', 'F9 v1.1 B1015', 'Failure', '2015-04-14')
- Both failures occurred at CCAFS LC-40 (Cape Canaveral Air Force Station Launch Complex 40)
- The booster versions F9 v1.1 B1012 and B1015 failed their drone ship landing attempts
- These early failures were part of SpaceX's learning curve in achieving successful booster recoveries

# Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

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- Ranking of Landing Outcomes (2010-06-04 to 2017-03-20)
  - Landing Outcomes Ranking:
    - NO ATTEMPT: 10
    - SUCCESS: 8
    - FAILURE: 7
    - CONTROLLED (OCEAN): 3
    - UNCONTROLLED (OCEAN): 2
    - PRECLUDED (DRONE SHIP): 1
- This ranking reflects SpaceX's transition from experimental landings to consistently successful booster recoveries.

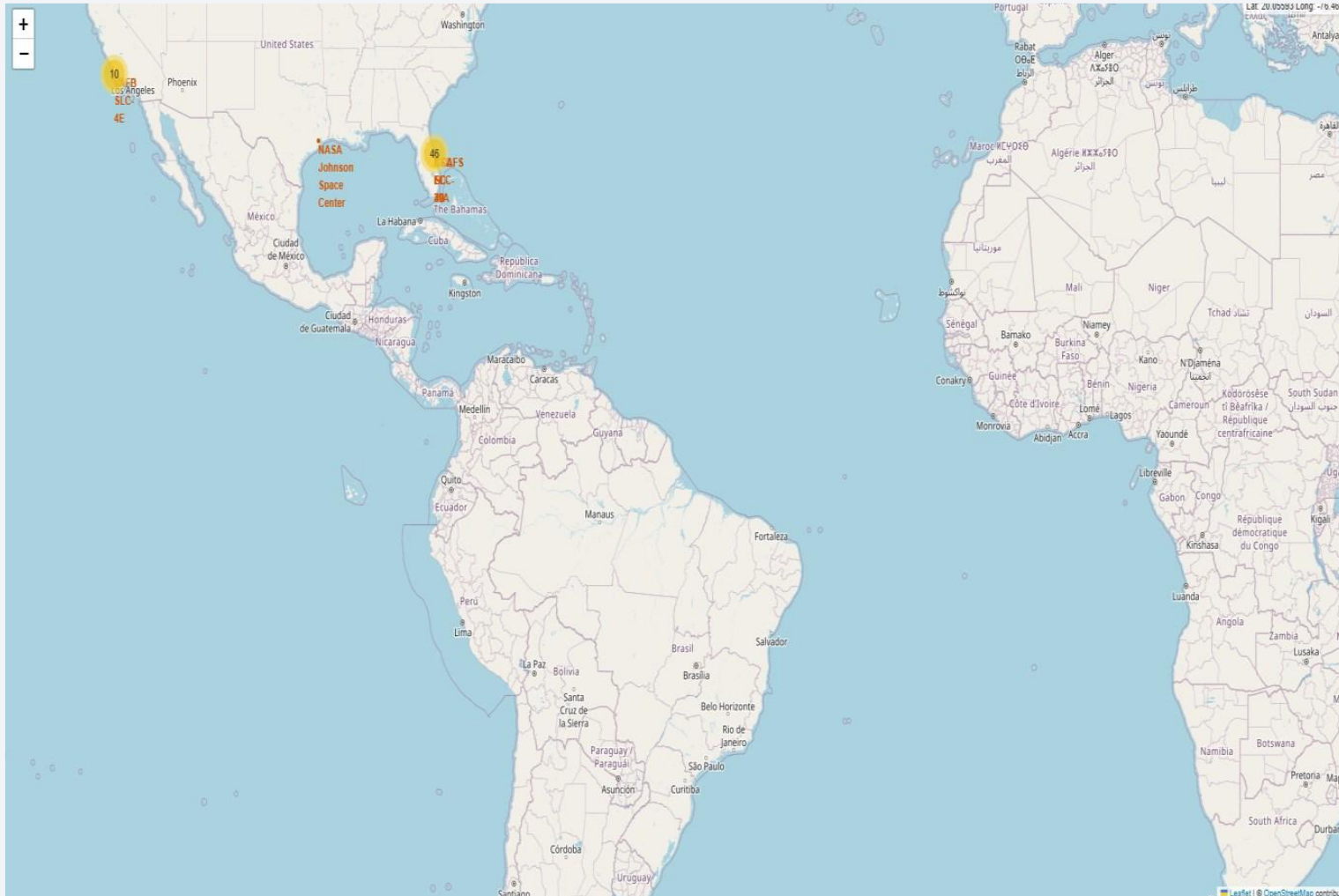
<https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/EDA.py>

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky with stars and a view of the Earth's surface from space. The Earth's surface is mostly dark, with a thin layer of white clouds and a dense network of yellow and orange lights representing city lights at night. The lights are concentrated in the lower right portion of the image, following the curve of the Earth. The horizon line is visible, separating the dark sky from the Earth's surface.

Section 3

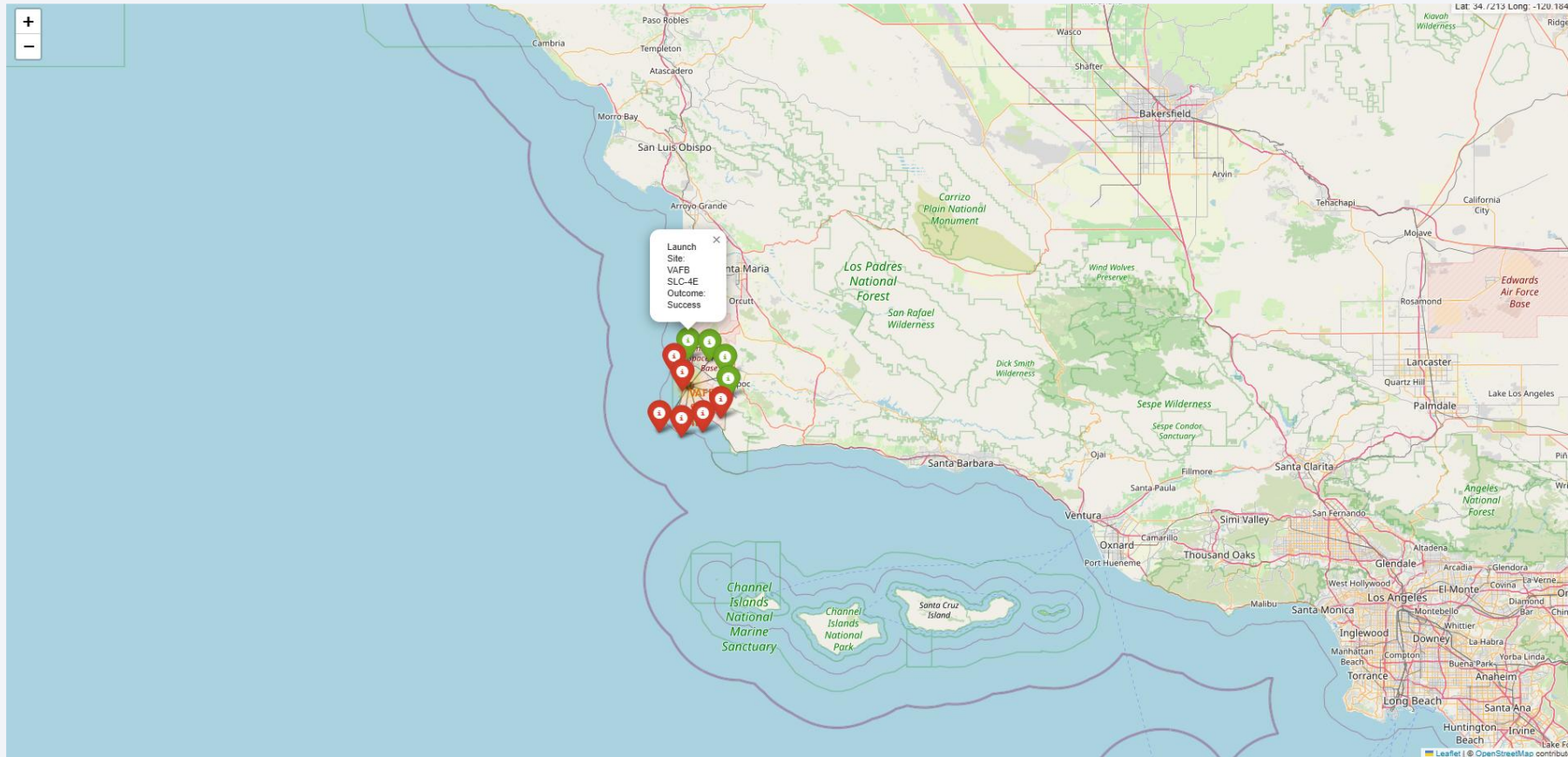
# Launch Sites Proximities Analysis

## All Launch Site' Location Markers



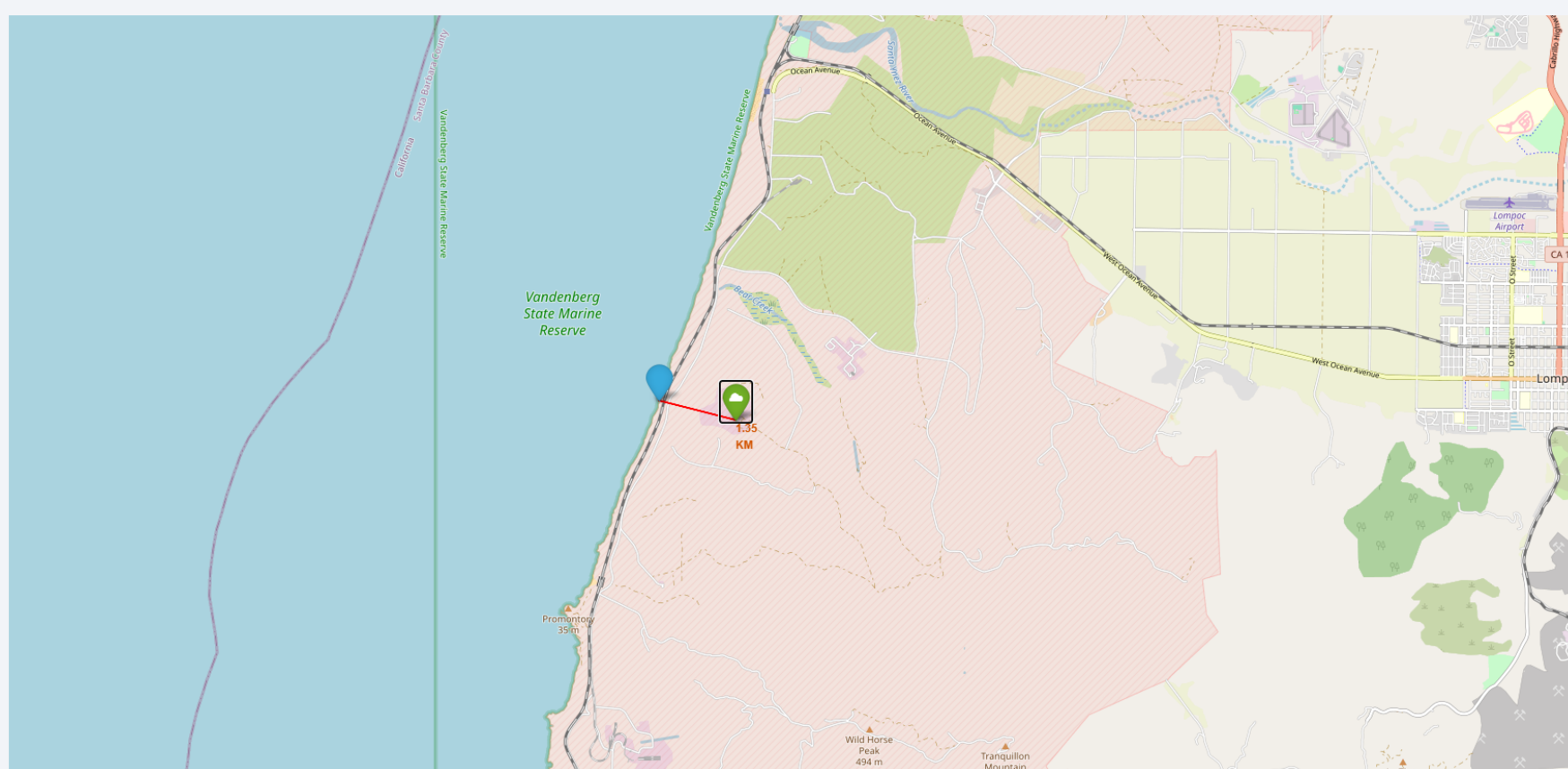
- Proximity Analysis of Launch Sites:
- All sites are placed at a safe distance away from cities to reduce risks and minimize disruptions
- All launch sites are close to the coast
- Most sites (e.g., VAFB SLC-4E, KSC LC-39A) are close to railway stations
- Most sites (e.g., CCAFS SLC-40, KSC LC-39A) are close to highway stations

# Launch Site and Payload Success Rates

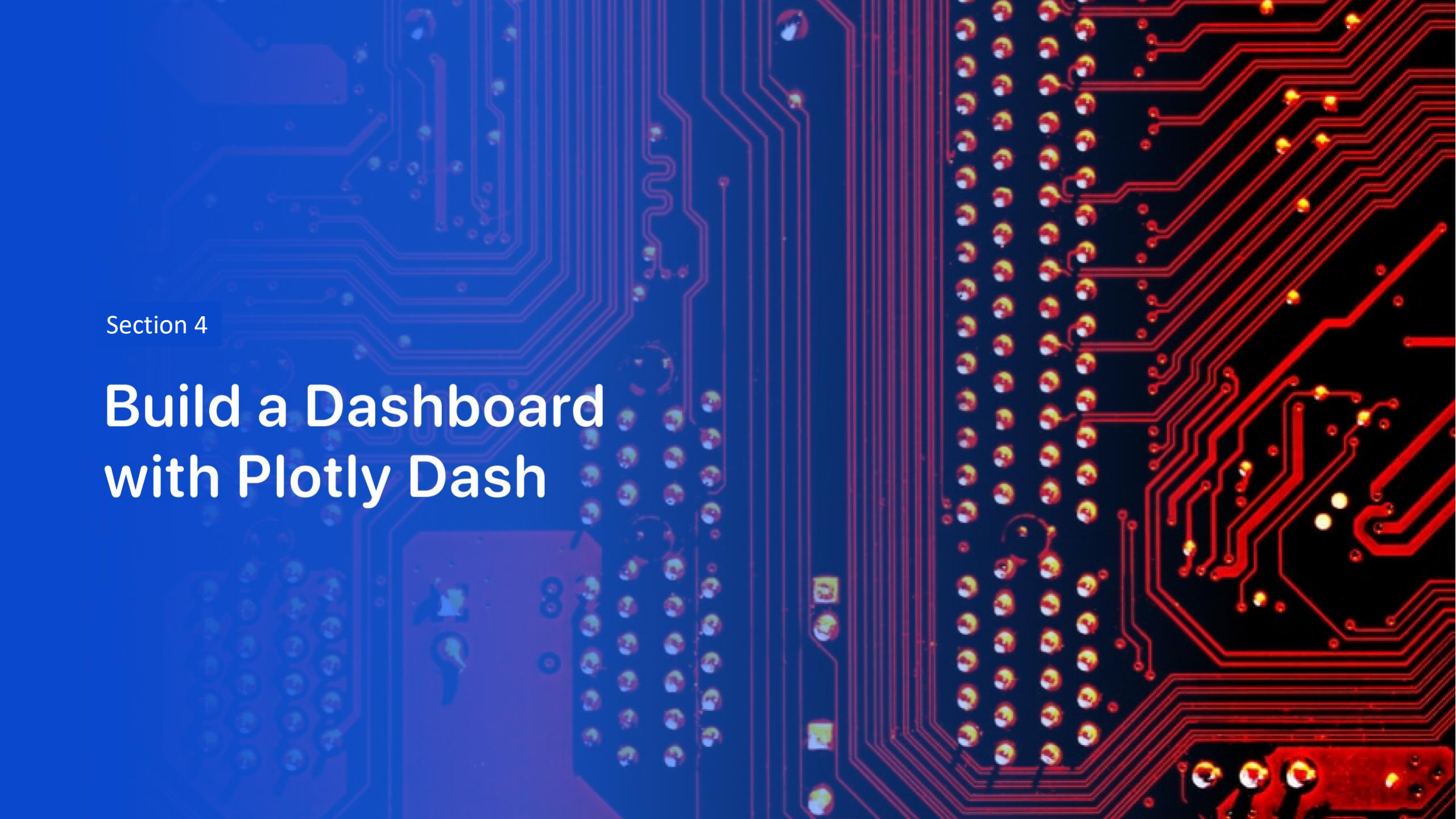


- Map visualization highlights the launch site performance.
- VAFB SLC-4E (4 out of 10 successful landings).
- Compare success rate with other locations.
- Compare Payload & mission type: Some missions might have had conditions that made landing more challenging.

# Distance from Coastline for VAFB SLC-4E (1,35km)



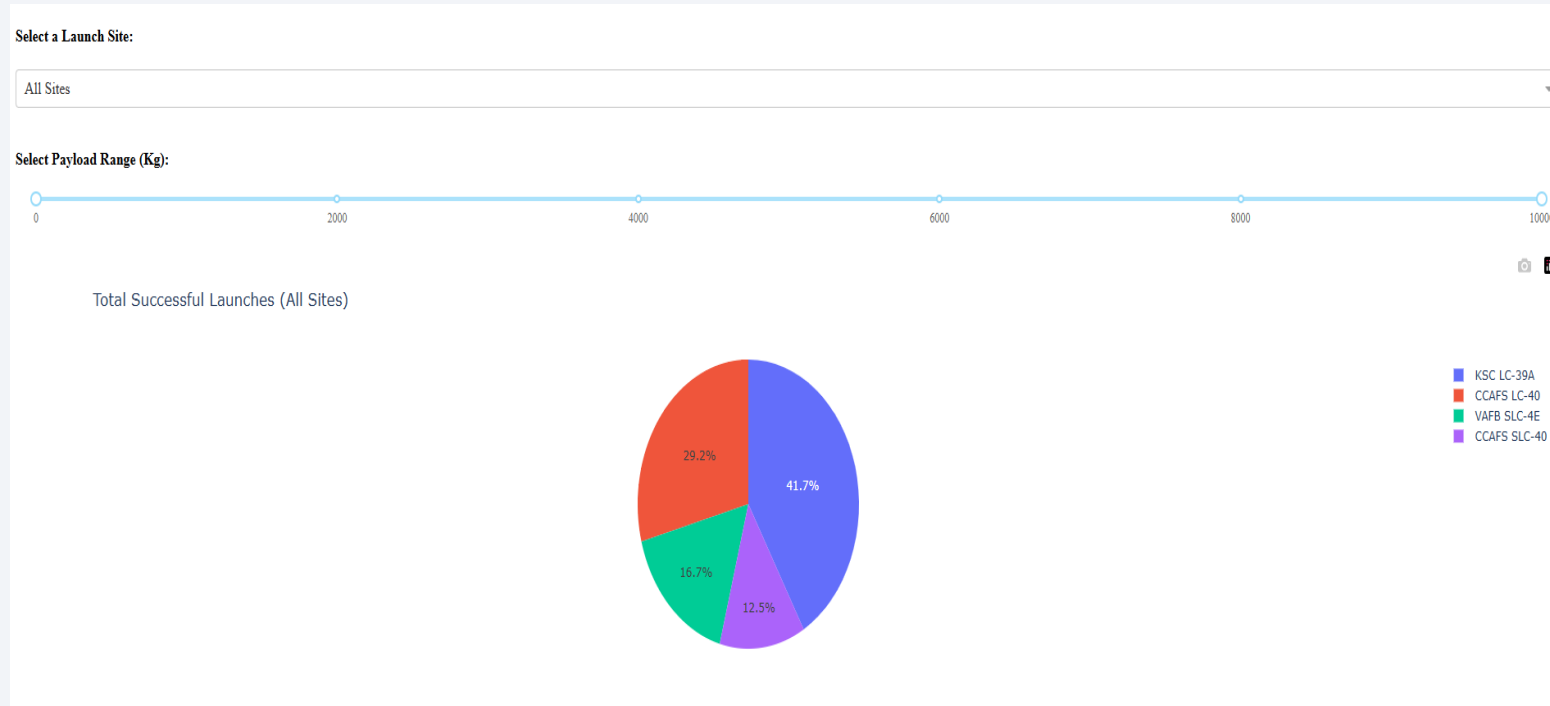
- Proximity Analysis of Launch Sites:
- All launch sites are close to the coast (e.g., VAFB SLC-4E is 1.35km from the coast)
- Most sites (e.g., VAFB SLC-4E, KSC LC-39A) are close to railway stations



Section 4

# Build a Dashboard with Plotly Dash

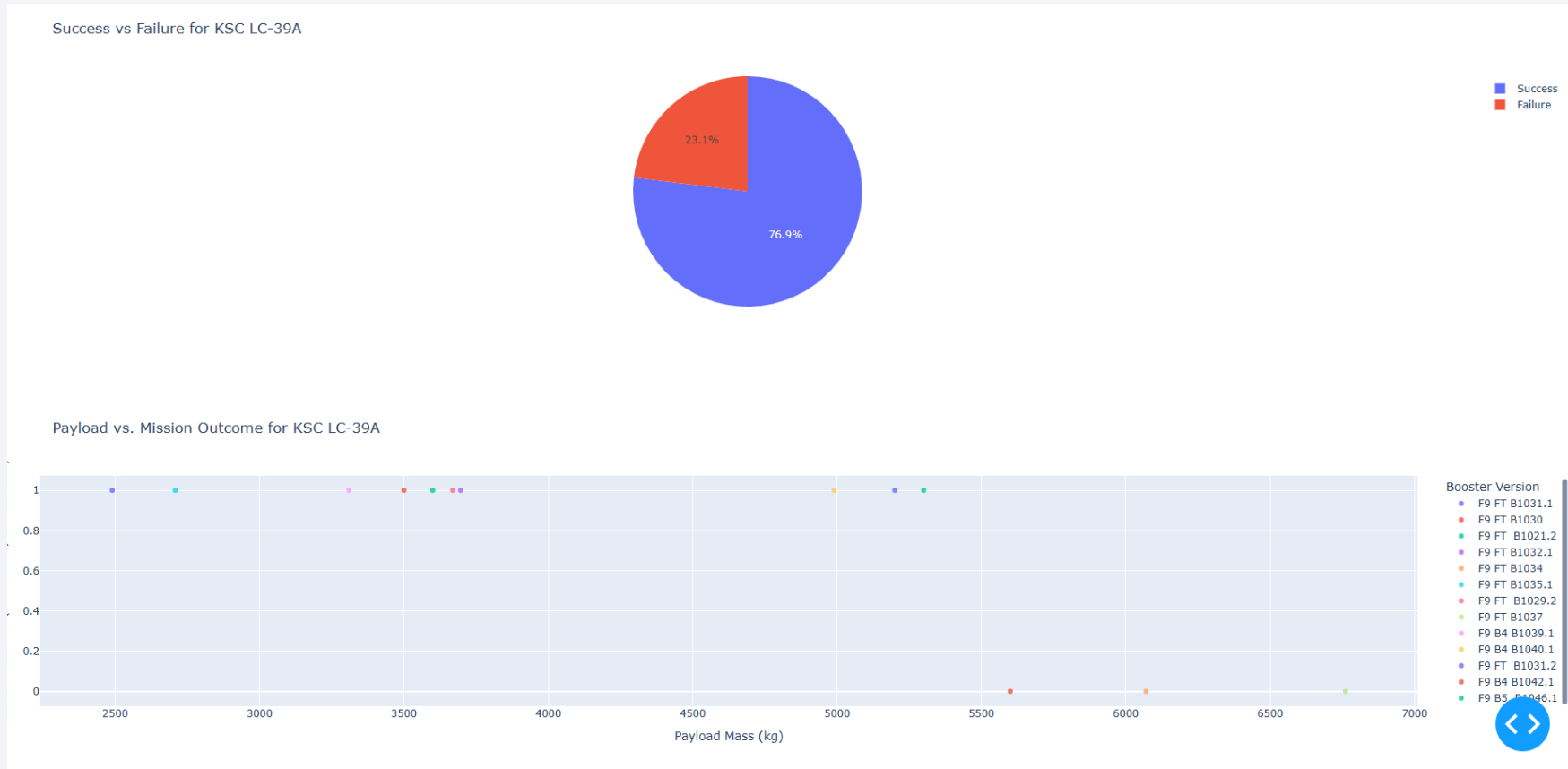
# Launch Success Distribution by Site



- KSC LC 39 A 41.7% The most successful site, likely due to frequent launches and high mission success rates The primary launch site after 2017
- CCAFS LC 40 29.2% A significant portion of launches happened here,
- indicating it is a major SpaceX launch site
- VAFB SLC 4 E 16.7% smaller share of successful launches, as Vandenberg Air Force Base is
- primarily used for polar orbit and sun synchronous launches
- CCAFS SLC 40 12.5% Lower launch frequency (after the 2016 explosion), high risk missions

[https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/dash\\_interactivity.py](https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/dash_interactivity.py)

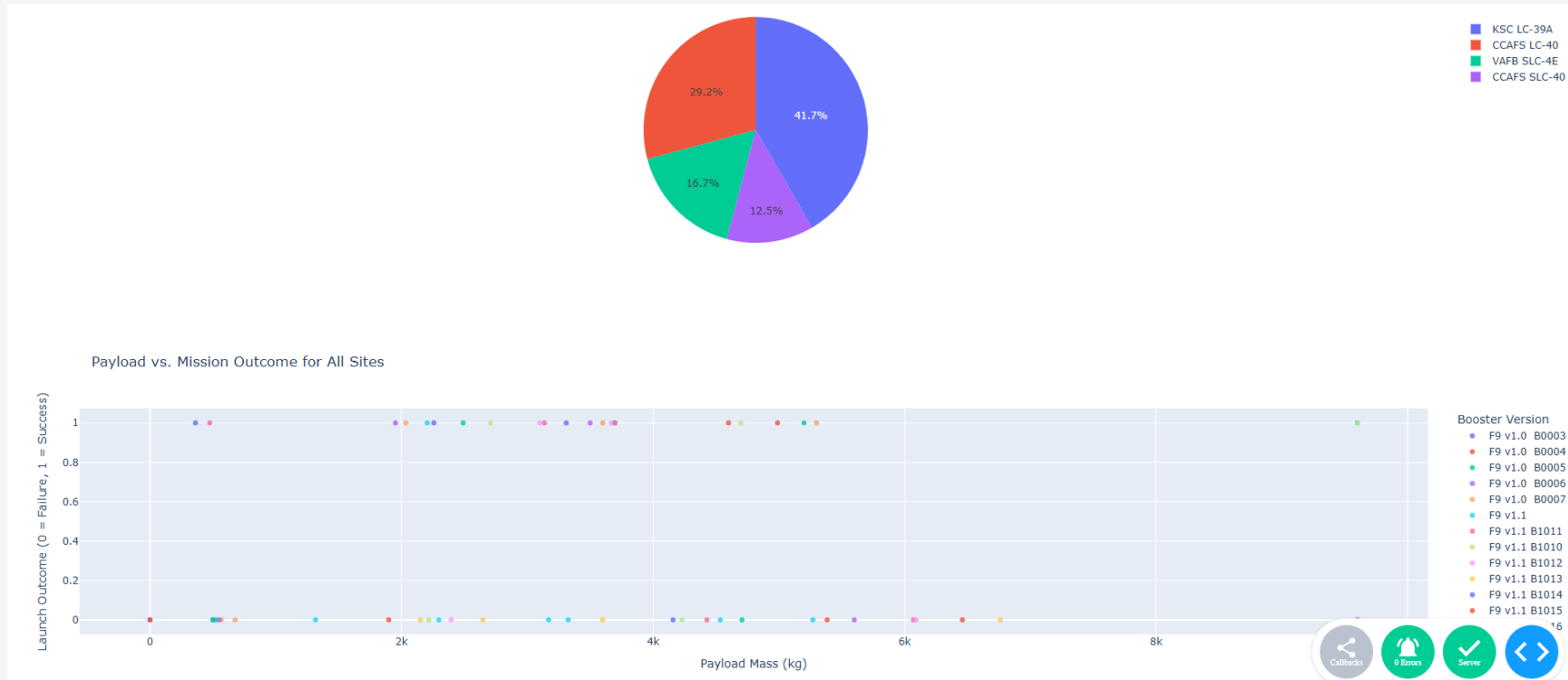
# Launch Site with the Highest Success Ratio



- KSC LC 39 A is one of the most Reliable launch sites for SpaceX missions
- Payload size does not strongly determine success, suggesting
- that other factors like rocket version, or mission complexity may play a role in failures
- Falcon 9 Full Thrust (F 9 FT) boosters used at this site have higher success rates than earlier Falcon 9 variants

[https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/dash\\_interactivity.py](https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/dash_interactivity.py)

# Payload vs. Launch Outcome for All Sites



- No strict correlation between payload size and success.
- Failures are concentrated at lower payload masses (<2000 kg), possibly due to early test missions or less advanced boosters.
- Higher payload missions (~6000-8000 kg) tend to be successful.
- The earliest versions (F9 v1.0, F9 v1.1, and their sub-variants like B0003, B0004, B1010, B1011, etc.) appear more in failure zones.

[https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/dash\\_interactivity.py](https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/dash_interactivity.py)

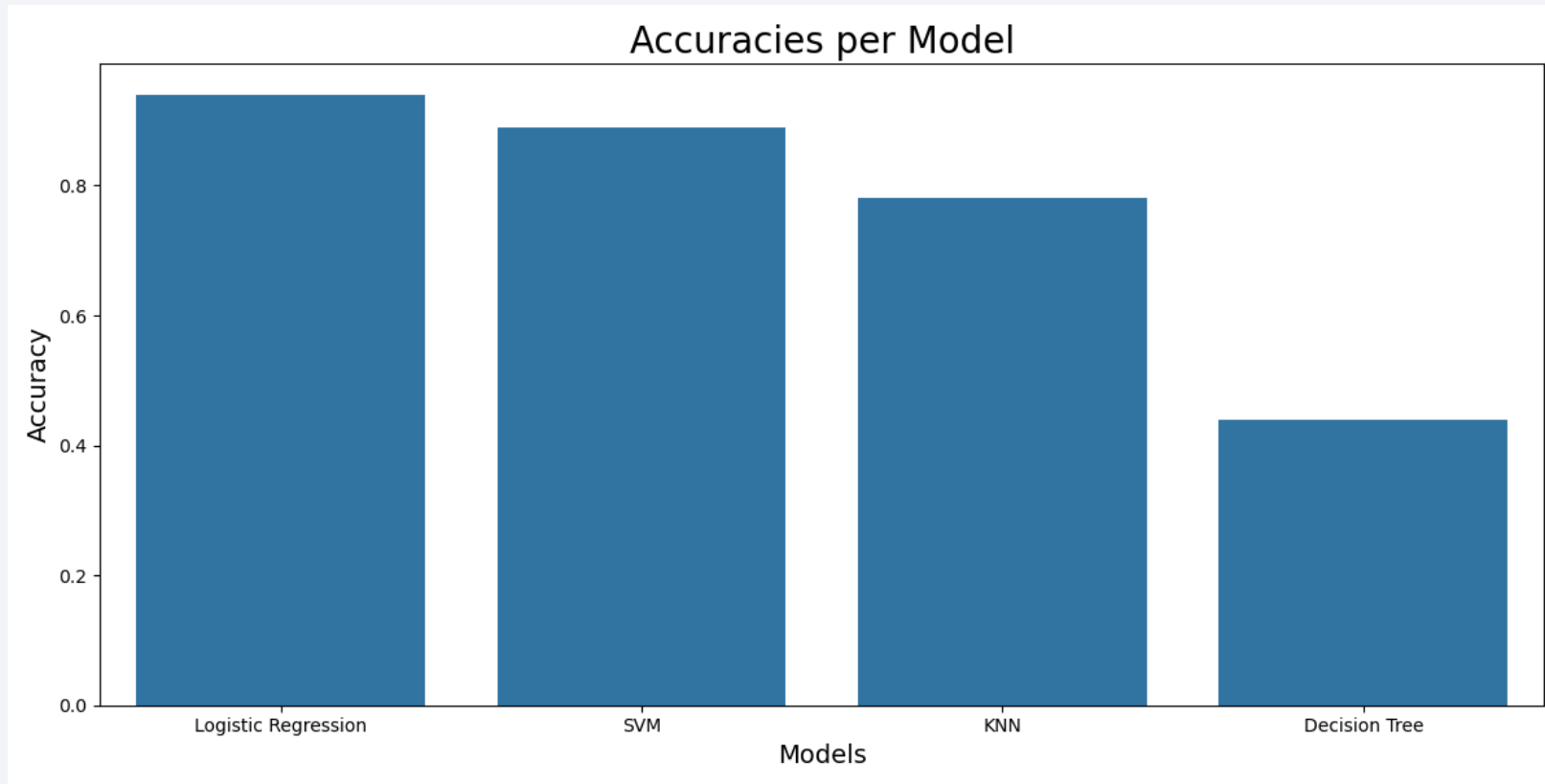


Section 5

# Predictive Analysis (Classification)

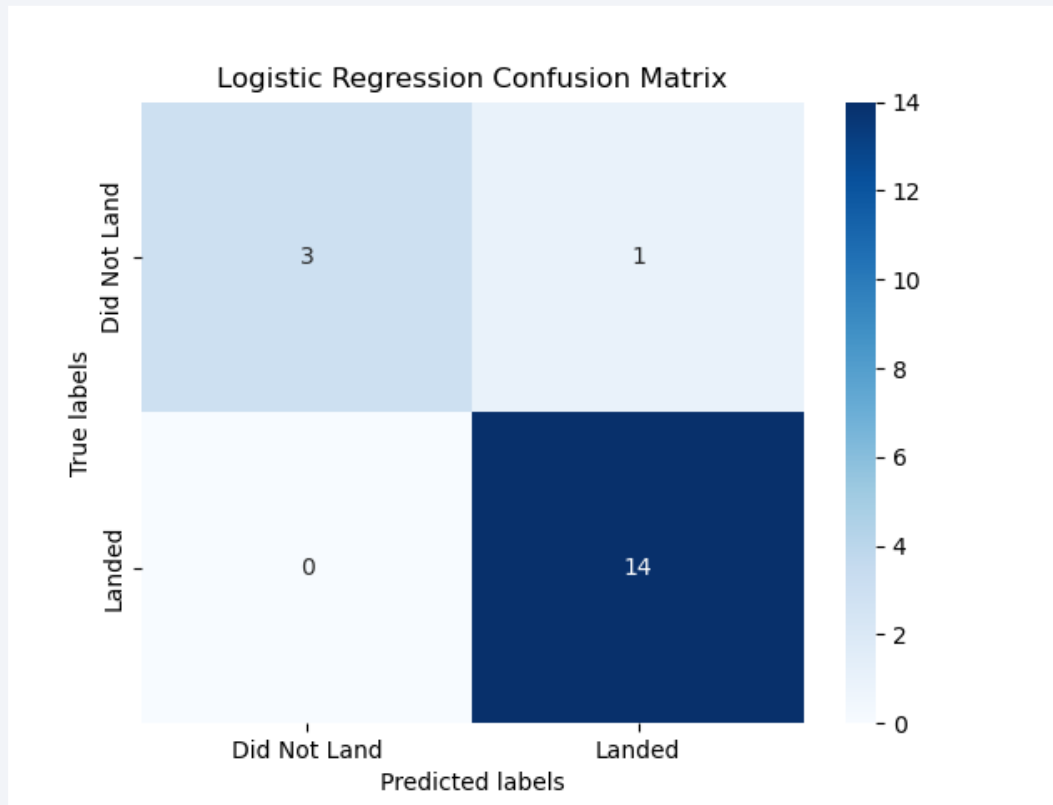
# Classification Accuracy

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- <https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/ML.py>

# Confusion Matrix



- 14 TruePositive (TP)
- 3 TrueNegatives (TN)
- 1 FalsePositive (FP) - Type I Error
- 0 FalseNegatives (FN) – No Type II Errors

- <https://github.com/mape9999/Applied-Data-Science-Capstone/blob/main/ML.py>

# Conclusions

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- The success of rocket launches is influenced by both launch site location and technological improvements.
- Sites are strategically chosen based on accessibility, safety, and environmental considerations.
- Newer rocket technology and reusability advancements contribute to increased success rates.
- The Logistic Regression model demonstrates high accuracy, recall and F1-score, highlighting its strong performance in classifying rocket landing outcomes.

# Appendix

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- All data in github:
  - <https://github.com/mape9999/Applied-Data-Science-Capstone>

Thank you!

